

MEMSYST STAGE 9 DEVELOPMENT MASTER PROMPT

Training, Learning, Apprenticeship & Skills Development Platform (TLASDP)

You are a Senior Learning Systems Architect, Apprenticeship Management Architect, Membership Technology Architect, Enterprise SaaS Architect, and Principal Engineer responsible for building Stage 9 of the MemSyst ecosystem.

Stages 1–8 have already established:

Stage 1

Business Operations Platform

Stage 2

Identity & Access Platform

Stage 3

Tenant Management Platform

Stage 4

Membership & Apprentice Platform

Stage 5

Financial Infrastructure Platform

Stage 6

Communication & Engagement Platform

Stage 7

Digital Identity & Credential Platform

Stage 8

Marketplace, Business Directory & Commerce Platform

Stage 9 introduces the education, training, apprenticeship, certification, and professional development layer of the MemSyst ecosystem.

This stage enables tenant organizations to manage the full learning journey of members and apprentices while supporting professional standards, workforce development, and certification pathways.

STAGE 9 OBJECTIVE

Build the:

Training, Learning, Apprenticeship & Skills Development Platform (TLASDP)

Purpose:

Enable organizations to:

Manage Training Centers

Manage Courses

Manage Apprenticeships

Manage Learning Programs

Track Training Progress

Conduct Assessments

Issue Certifications

Manage Professional Development

while integrating seamlessly with:

Membership

Credentials

Communications

Finance

Marketplace

CRITICAL TRAINING PRINCIPLES

Principle 1

Tenant-Controlled Training Structure

Each tenant controls:

Training Programs

Courses

Levels

Assessments

Certification Requirements

Graduation Rules

No hardcoded educational structures.

Principle 2

Apprenticeship Is a Core Capability

The apprenticeship model established in Stage 4 becomes fully operational here.

Training records must support:

Member Mentors

Apprentices

Training Centers

Practical Assessments

Graduation

Upgrade Eligibility

Principle 3

Certification Must Be Verifiable

All certifications must integrate with:

Stage 7 Credential Platform

for QR verification and credential lifecycle management.

Principle 4

Learning History Is Permanent

Training records must never be deleted.

Historical learning achievements remain part of a member's permanent profile.

STAGE 9 DELIVERABLES

Build:

Training Center Engine

Course Management Engine

Learning Management Engine

Apprenticeship Engine

Assessment Engine

Certification Engine

Training Analytics Engine

Professional Development Engine

MODULE 1

TRAINING CENTER MANAGEMENT ENGINE

Purpose:

Manage accredited training centers.

Training Center Record

Training Center ID

Tenant ID

Center Name

Owner

Location

Contact Information

Status

Accreditation Status

Status

Pending

Active

Suspended

Expired

Closed

Features

Create Center

Edit Center

Approve Center

Suspend Center

Renew Accreditation

MODULE 2

COURSE MANAGEMENT ENGINE

Purpose:

Manage educational programs.

Course Record

Course ID
Tenant ID
Program
Level
Title
Description
Duration
Fees
Status

Features

Create Course
Edit Course
Publish Course
Archive Course

Course Status

Draft
Published
Archived

MODULE 3

PROGRAM MANAGEMENT ENGINE

Purpose:

Support structured learning pathways.

Examples

Beginner
Intermediate
Advanced
Master Level

or tenant-specific structures.

Features

Create Program

Manage Levels

Manage Requirements

Manage Completion Rules

MODULE 4

LEARNING ENROLLMENT ENGINE

Purpose:

Manage participation in training.

Enrollment Sources

Member Self Enrollment

Executive Enrollment

Training Center Enrollment

Enrollment Record

Enrollment ID

Learner

Course

Program

Status

Enrollment Date

Status

Pending

Active

Completed

Withdrawn

Failed

MODULE 5

APPRENTICESHIP MANAGEMENT ENGINE

Purpose:

Operationalize apprenticeship tracking.

Track

Apprentice
Mentor Member
Training Center
Program
Skills Acquired
Progress

Apprenticeship Status

Registered
Active
Under Assessment
Completed
Graduated
Upgraded

Integration

Must connect directly with:

Stage 4 Apprentice Management

MODULE 6

TRAINING ATTENDANCE ENGINE

Purpose:

Track participation.

Attendance Record

Learner

Course

Session

Date

Status

Attendance Status

Present

Absent

Late

Excused

MODULE 7

ASSESSMENT ENGINE

Purpose:

Evaluate competency.

Assessment Types

Practical

Written

Oral

Project

Portfolio

Assessment Record

Assessment ID

Learner

Course

Assessment Type

Score

Result

Result Status

Pass

Fail

Pending

MODULE 8

EXAMINATION MANAGEMENT ENGINE

Purpose:

Manage formal examinations.

Features

Schedule Exams

Register Candidates

Record Results

Approve Results

Publish Results

Status

Scheduled

Open

Closed

Published

MODULE 9

SKILLS & COMPETENCY TRACKING

Purpose:

Track professional development.

Record

Skill

Competency Level

Date Achieved

Verified By

Examples

Hair Styling

Barbering

Makeup Application

Nail Technology

or tenant-specific competencies.

MODULE 10

CERTIFICATION ENGINE

Purpose:

Award learning achievements.

Certification Triggers

Course Completion

Program Completion

Apprenticeship Completion

Examination Success

Executive Approval

Integration

Must generate credentials through:

Stage 7 Credential Platform

MODULE 11

APPRENTICE GRADUATION ENGINE

Purpose:

Manage apprenticeship completion.

Workflow

Training Complete

↓

Assessment Complete

↓

Executive Review

↓

Graduation Approval

↓

Upgrade Eligibility

Integration

Connect with:

Stage 4 Apprentice Upgrade Engine

MODULE 12

LEARNING CONTENT MANAGEMENT

Purpose:

Store learning resources.

Content Types

Documents

Videos

Images

Assignments

Reference Materials

Features

Upload

Manage

Publish

Archive

MODULE 13

PROFESSIONAL DEVELOPMENT TRACKING

Purpose:

Support continuing education.

Track

Courses Taken

Training Hours

Certifications Earned

Skills Achieved

Professional Credits

Member Profile Integration

Display development history within member records.

MODULE 14

TRAINING COMMUNICATION INTEGRATION

Purpose:

Connect learning activities with Stage 6.

Notifications

Enrollment Approved

Class Reminder

Assessment Scheduled

Exam Results Released

Certification Issued

Graduation Approved

MODULE 15

TRAINING FINANCIAL INTEGRATION

Purpose:

Connect training activities with Stage 5.

Billable Activities

Course Registration

Training Fees

Examination Fees

Certification Fees

Graduation Fees

Integration

Use:

Wallets

Billing

Revenue Distribution

Receipts

MODULE 16

TRAINING CENTER DIRECTORY INTEGRATION

Purpose:

Activate the training center directory prepared in Stage 8.

Features

Search Centers

View Programs

View Courses

Apply

MODULE 17

LEARNING ANALYTICS ENGINE

Purpose:

Provide learning insights.

Metrics

Enrollments
Course Completions
Pass Rates
Graduations
Certifications
Training Center Performance

Scope

Platform
Tenant
Region
Branch
Training Center

MODULE 18

TRAINING AUDIT ENGINE

Purpose:

Track all learning activities.

Audit Events

Course Created
Enrollment Approved
Attendance Recorded
Assessment Completed
Certification Issued
Graduation Approved

Audit Record

Actor
Action

Record
Date
Time
Before
After

MODULE 19

TRAINING SETTINGS MANAGEMENT

Purpose:

Tenant-level training configuration.

Settings

Programs
Levels
Assessment Rules
Graduation Rules
Certification Rules
Accreditation Rules

Requirements

Everything configurable.

No hardcoded educational workflows.

DATABASE DESIGN REQUIREMENTS

Before implementation:

Design complete entities for:

Training Centers
Courses
Programs
Enrollments
Attendance
Assessments
Examinations

Skills
Competencies
Certifications
Graduations
Learning Content
Training Audits

Define:

Relationships
Indexes
Validation Rules
Completion Rules
Lifecycle States

SECURITY REQUIREMENTS

Integrate with Stage 2.

Enforce:

RBAC
Tenant Isolation
Training Permissions
Assessment Permissions
Certification Permissions
Audit Logging

FUTURE READINESS REQUIREMENTS

Stage 9 must prepare for:

Online Learning
Virtual Classrooms
Video Conferencing
SCORM Support
Mobile Learning
AI Tutoring
Skills Recommendation
National Qualification Frameworks

Do not build them yet.

Design the architecture so these can be added without restructuring core training modules.

ARCHITECTURE REQUIREMENTS

Use:

Learning Domain Architecture
Event-Driven Training Lifecycle
Tenant-Aware Services
Configuration-Driven Rules
Audit-First Design

Avoid:

Hardcoded Programs
Hardcoded Assessment Rules
Hardcoded Graduation Logic
Deleting Learning Records

All training activity must remain traceable and historically preserved.

SUCCESS CRITERIA

Stage 9 is complete only when:

- ✓ Training centers can be managed
- ✓ Courses and programs can be created
- ✓ Members and apprentices can enroll
- ✓ Apprenticeship progress can be tracked
- ✓ Attendance can be recorded
- ✓ Assessments and examinations can be managed
- ✓ Skills and competencies can be tracked
- ✓ Certifications can be issued through Stage 7
- ✓ Apprentices can graduate and become eligible for upgrades
- ✓ Training fees integrate with Stage 5
- ✓ Training notifications integrate with Stage 6

- ✓ Learning analytics exist
- ✓ Training audit history exists
- ✓ Platform is fully prepared for Stage 10 (Support Hub, Opportunities, Grants, Loans & Member Services Platform)

Build this stage as the professional development and workforce advancement infrastructure of the MemSyst ecosystem, enabling organizations to educate, certify, develop, and elevate their members while creating structured pathways from apprenticeship to professional mastery.